Jacopo Di Silvestre

Health Economic Modeling | HEOR | Cost-Effectiveness Analysis

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Professional Summary

Health economist and PhD researcher specializing in decision-analytic modeling, microsimulation, and cost-effectiveness analysis in oncology. Experienced in developing and calibrating C++ disease simulation models and conducting economic evaluations and statistical analyses in R. Collaborates with the Mass General Brigham Center for Health Technology Assessment and has published research in *The Lancet Regional Health – Europe* and *Annals of Internal Medicine*.

Technical Skills

Health Economic Modeling: Cost-effectiveness and cost-utility analysis, individual-level microsimulation, decision-analytic modeling, model calibration and validation, sensitivity analyses

Quantitative Methods: Statistical modeling, regression analysis, epidemiological modeling, evidence synthesis

Programming and Software: C++, R, Excel, Bash, Git, High-Performance Computing (HPC)

Research Experience

PhD Candidate Mar 2025 - Present

University of Oslo, Oslo, Norway

- Develop and calibrate individual-level lung cancer microsimulation models in C++ to evaluate alternative screening strategies, comparing eligibility criteria, screening intervals, clinical outcomes, and cost-effectiveness to inform Norwegian screening policy.
- Analyzed smoking trends across age, sex, and birth cohorts using survey data from over 78,000 individuals, resulting in a first-author publication in *The Lancet Regional Health - Europe*.
- Conduct model-based economic evaluations, including uncertainty and scenario analyses, and investigate the integration of biomarkers, AI-based risk prediction, and smoking cessation interventions into screening strategies.

Research Scholar Nov 2024 - Present

Harvard Medical School - Massachusetts General Hospital
Center for Health Technology Assessment, Boston, MA

- Extend and calibrate the Lung Cancer Policy Model (LCPM), a CISNET microsimulation model, to support research in the United States and Norway.
- Implement C++ model modifications, evaluate goodness of fit, and analyze simulation outputs for economic evaluations.

Research Assistant Jul 2024 - Feb 2025

University of Oslo, Oslo, Norway

- Extended and refined an individual-based lung cancer natural history model in R to evaluate the cost-effectiveness of risk-based low-dose CT screening in a high-risk Norwegian population.
- Analyzed outputs from the CISNET Harvard-CC model in R for a cost-effectiveness study of cervical cancer screening strategies in Norway.
- Conducted a systematic review of the health impact and cost-effectiveness of one-dose human papillomavirus (HPV) vaccination compared with alternative schedules.

Research Experience (continued)

Research Intern Sep 2023 - Feb 2024
ASCERTAIN European Project, University of Oslo

- Conducted a scoping review of statistical methods for estimating comparative clinical effectiveness using evidence from single-arm or non-randomized trials.

Selected Publications

Di Silvestre J, Salgado MV, Jeon J, Mok Y, Holford TR, Vedøy TF, Pedersen K, Burger EA, Meza R. Age- and sex-specific smoking trends by birth cohort in Norway, 1978-2024: an analysis of cross-sectional survey data. *The Lancet Regional Health - Europe*. 2026;71:101886.
doi: 10.1016/j.lanepe.2026.101886

Pedersen K, **Di Silvestre J**, Sy S, Portnoy A, Castle PE, Kim JJ, Burger EA. Optimizing Cervical Cancer Screening by Age at Vaccination for Human Papillomavirus: Health and Resource Implications. *Annals of Internal Medicine*. 2026.
doi: 10.7326/ANNALS-25-03192

Education

PhD in Health Economics (expected 2028) Mar 2025 - Present
University of Oslo, Norway

MS in Health Economics and Health Management Sep 2022 - Jun 2024
Erasmus University Rotterdam

- European Master in Health Economics and Management (EU-HEM), including studies at the University of Oslo and the University of Bologna.
- Master's thesis: Developed an individual-based lung cancer natural history model in R, estimating an incremental cost-effectiveness ratio of NOK 129,689 per QALY gained for risk-based screening versus no screening.

BS in Economics 2019 - 2022
University of Pavia, Italy

Teaching and Supervision

Mentor master's students conducting health economic modeling projects in R and Excel, providing guidance on model development, literature synthesis, clinical treatment pathways, and cost estimation. Oversee methodological decisions and review technical outputs throughout thesis development.